Guaranteed navigation in unknown off-road environments

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Context

Outdoor unknown off-road navigation stakes in mobile robotics applications are multiple:

- The need to specify **ground geometry**, its nature (e.g., slippery ground), as well as the presence and nature of **dynamic objects** to adjust the mapping models and potentially to account for their avoidance in path generation.
- **Changes in the environment** (variations in luminosity, rain, fog, etc) do not always allow the continuous use of the same sensors. Therefore, sensors selection but also multi-sensor fusion, is essential.
- The **noisy nature of sensor acquisitions** must be the subject to careful consideration to provide robust information (possibly guaranteed using Interval Analysis [1]) on which path generation can be based. The context of the described environment corresponds to applications such as construction or farming sites where vehicles evolve jointly with human operators, applications where safety is the top priority.

A robust sensor: The micro-wave RADAR

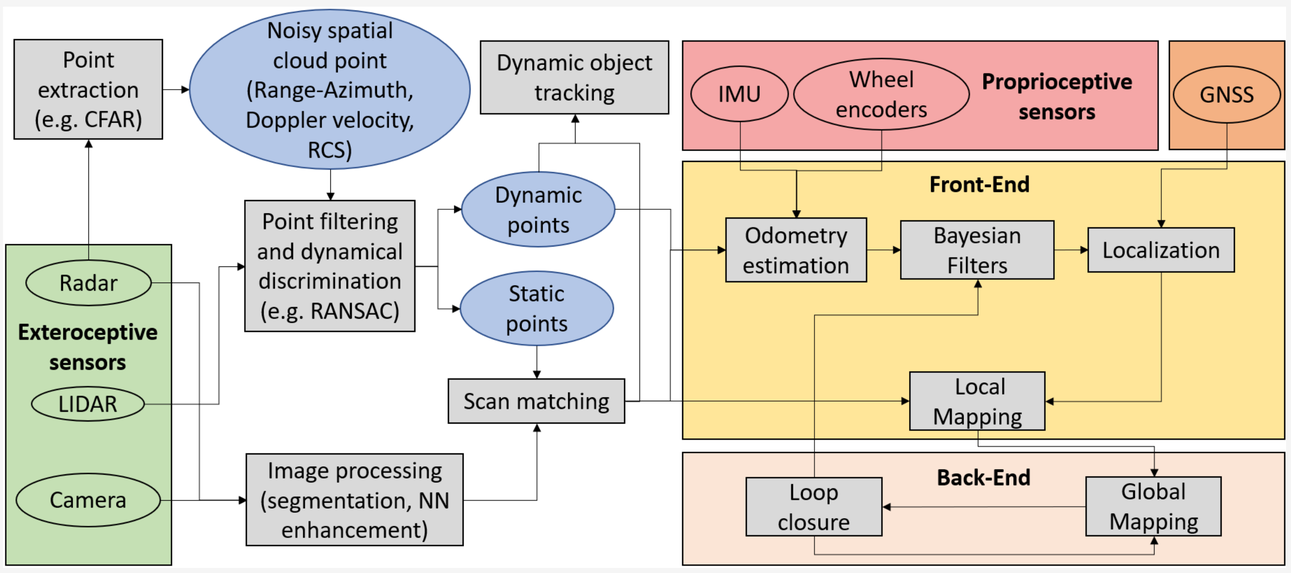

Figure 1. Overview of radar integration in SLAM.

Similar to LiDAR, RADAR is an electromagnetic-based active sensor, that returns the distance to an obstacle using the time of flight between signal transmission and reception. The difference is that the wave frequencies lie in the radio-frequency range, between 3 MHz and 300 GHz (76 - 81 GHz for automotive RADAR), whereas LiDAR wave frequency operates in the infrared range, at approximately 10,000 times higher frequencies. This difference in wave frequency translates into sensor property where RADAR:

- Is less sensitive to signal attenuation in the atmosphere, allowing **higher maximum range and greater robustness** to weather conditions such as rain or fog.
- Returns the **Doppler velocity** of obstacles.
- Can potentially **penetrate certain materials** depending on the type of RADAR.
- Is **cheaper** than LiDAR, making it attractive for industrial applications.

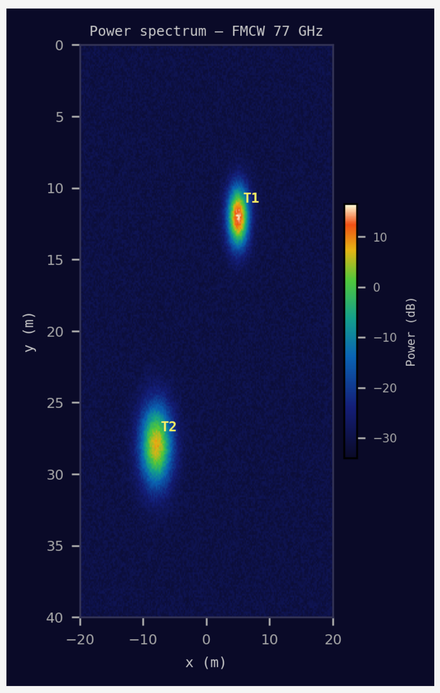

Figure 2. Example of RADAR output.

However, RADAR also has some drawbacks when compared to LiDAR:

- RADAR has a **lower detection accuracy** compared to LiDAR requiring more careful noise modelling.
- Its **low spatial resolution** makes it complicated to detect weak Radar Cross Section target (RCS) depending on shape, geometry and material.

So, although RADAR appears to be a cheap and robust sensor for harsh environmental conditions, it is difficult to use without signal processing and/or without combining it with other sensors [2].

An application of the radar benefit: The guaranteed dynamic tracking

Dynamic objects in the environment need to be identified in order to

- Correctly update the map when changes occur.
- Avoid dynamic obstacles.

Dynamic obstacles are of two types:

- Semi-static objects whose positions do not change frequently (e.g., a door).
- Highly dynamic objects which move frequently (e.g., a person).

Classically, when range-azimuth sensors are used, dynamic object tracking is performed by comparing multiple frames in order to detect both semi-static and highly dynamic behaviours. When Doppler velocity is available, instantaneous velocity, including angular speed, can be estimated with the Instantaneous Center of Rotation (ICR) [3] allowing to predict the positions of measured points at the next time and update a geometrical map. These kinematic parameters can be estimated using Interval Analysis taking into account bounded RADAR noise.

$$y_r \cos(\theta_i) - x_r \sin(\theta_i) = y_{ICR} \cos(\theta_i) - x_{ICR} \sin(\theta_i) - v_{d_i} \cdot \omega^{-1} \quad (1)$$

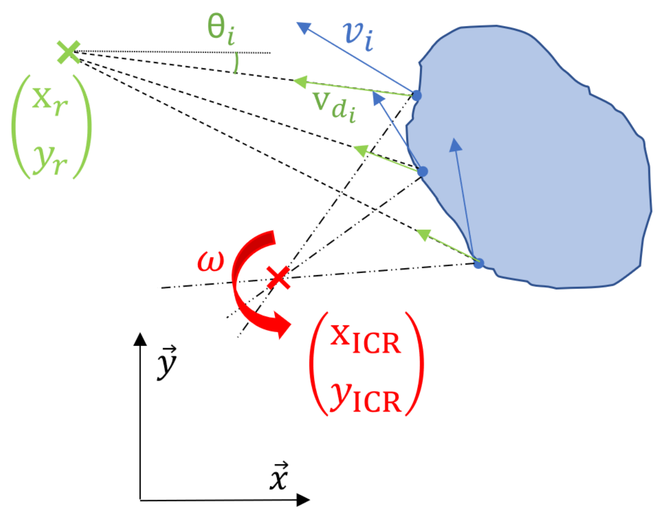

Figure 3. ICR and angular velocity estimation using of Doppler radial velocity distribution.

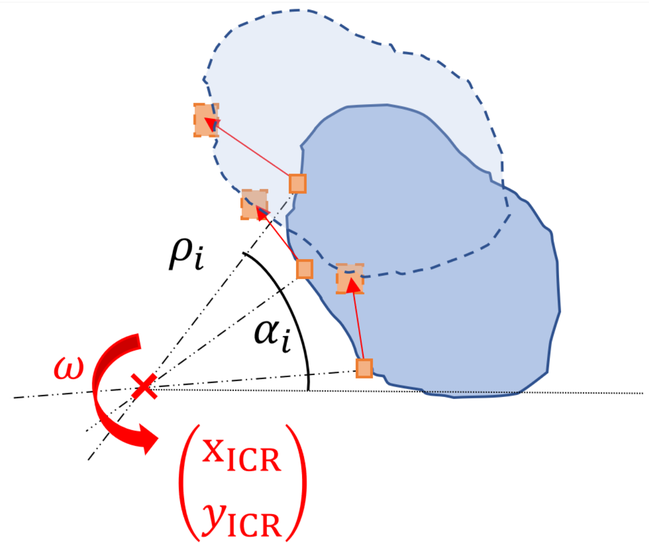

Figure 4. Projection of measured points with ICR and angular velocity estimation.

Research tracks

- How to estimate dynamic object shapes in a guaranteed way without making shape assumptions?
- What can be concluded from the observation of dynamic object trajectories inside a static map (e.g. Visibility, non traversability of objects)?

Bibliography

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[3] Dominik Kellner, Michael Barjenbruch et Jens Klappstein. "Tracking of extended objects with high-resolution Doppler radar". In : *IEEE Transactions on Intelligent Transportation Systems* 17.6 (2015), p. 1341-1353.